

The Alan Machine

João Alisson and contributors

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Preface

For Alan Turing, who imagined the machine,
and for Alan Ricardo.

How to read this book

This book is about a machine that does not exist. We call it Alan, and we use it to ask a question physics takes very seriously: can entropy be reversed, and if it could, what would happen to time?

You do not need a physics degree to read it. The chapters are written for anyone who is curious. When the math helps, it shows up in boxes marked *Going deeper*, which you can open or skip. The story works either way.

Some chapters have a companion paper, written for specialists, with the full argument. You will find the papers near the end of the book, and each chapter tells you when one exists.

Each part ends with an *interlude*: a short pause for a question that physics can inform but not answer, such as whether time really flows or whether a machine can think. After the main text, *Building Alan* looks at real technology, from today's chips to quantum computers, and asks how far each one is from the limits the chapters describe. Those pages are updated every year.

Along the way, we keep four kinds of statement apart. Unmarked claims are **established** physics, with references you can check. Boxes marked **Extrapolation** follow from established physics, but only if their stated assumptions hold. Boxes marked **Speculation** hold ideas that are conceivable but that no evidence supports yet. Boxes marked **Philosophy** hold questions that physics cannot settle, with the best arguments on each side. A thought experiment is only useful if you always know which of the four you are reading.

Every technical term links to its plain-language definition in the Concepts appendix at the end.

Part I

The Machine and the Question

1 Meet Alan

This chapter has not been written yet.

2 What Entropy Is

This chapter has not been written yet.

3 The Arrow of Time

This chapter has not been written yet.

Interlude: Does Time Flow?

This interlude has not been written yet.

Part II

Information Is Physical

4 Maxwell's Demon and the Szilard Engine

This chapter has not been written yet.

5 Landauer's Principle

This chapter has not been written yet.

6 Reversible Computing

This chapter has not been written yet.

7 Can a Mind Beat the Second Law?

This chapter has not been written yet.

Interlude: Three Demons

This interlude has not been written yet.

Part III

The Limits of the Machine

8 How Fast Can It Compute

This chapter has not been written yet.

9 How Much Can It Hold

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10 Is Alan a Quantum Computer?

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11 Laplace's Demon

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12 Black Holes and Information

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Interlude: Are We Inside an Alan?

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Part IV

Reversing Entropy

13 Local Reversal

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14 Life: The Original Local Reversal

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15 Fluctuations and Recurrence

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16 Global Reversal

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Interlude: Would You Remember?

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Part V

The Impact on Time

17 What Time Is

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18 Would Time Run Backward

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19 Closed Timelike Curves

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20 The End of the Universe

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Interlude: Can Alan Think?

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Epilogue: What Alan Cannot Do

This chapter has not been written yet.

Part VI

Building Alan

About Building Alan

The chapters ask what physics allows. This part asks a different question: how close does today's technology come, and what would close the gap? The answer changes every year, so these pages are living documents. Each one has a review date, and every number in it carries the date it refers to and its source.

Each dossier looks at one technology or one bottleneck and measures it against the same yardsticks:

- **The Landauer bound:** the least heat that erasing one bit can release, $k_B T \ln 2$ at the temperature the device runs at (Landauer 1961; Bérut et al. 2012).
- **The Margolus-Levitin bound:** the most operations per second that a given amount of energy allows (Margolus and Levitin 1998; Lloyd 2000).
- **The Bekenstein bound:** the most information that a given mass and size can hold (Bekenstein 1981).
- **Energy at the wall plug:** what the whole system draws from the grid, cooling and power delivery included. A device that runs near the Landauer bound at 4 K can still cost more, counted at the wall plug, than one at room temperature.

These pages are written for engineers and researchers who design processors, memory, AI hardware and data centers, and for any reader who wants to know where real machines stand. The numbers behind them are published as data (see [the data folder](#)), so they can be reused and checked.

The same method, applied beyond computing, is [Escape Velocity](#): an open atlas of what each technology still needs to reach maturity, from fusion to medicine, and of which technologies wait on which. Its entries on quantum computers, data movement, heat removal and the energy of AI inference link back to these dossiers.

Dossiers

Dossier	Status	Last reviewed	Review due	Curators
CMOS: The Baseline	proposed	never	not reviewed yet	none yet

Dossier	Status	Last reviewed	Review due	Curators
The Cost of Moving Data	proposed	never	not reviewed yet	none yet
Getting the Heat Out	proposed	never	not reviewed yet	none yet
Adiabatic and Reversible Logic	proposed	never	not reviewed yet	none yet
Superconducting Logic	proposed	never	not reviewed yet	none yet
Quantum Hardware	proposed	never	not reviewed yet	none yet
Thermodynamic and Probabilistic Computing	proposed	never	not reviewed yet	none yet
Neuromorphic Computing	proposed	never	not reviewed yet	none yet
Optical Computing	proposed	never	not reviewed yet	none yet
The Energy of a Token	proposed	never	not reviewed yet	none yet

CMOS: The Baseline

This dossier has not been written yet.

The Cost of Moving Data

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Getting the Heat Out

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Adiabatic and Reversible Logic

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Quantum Hardware

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Optical Computing

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The Energy of a Token

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Part VII

Papers

About the Papers

The chapters tell the story. The papers make the argument.

Some chapters have a companion paper with the full scientific treatment of their subject: the derivations, the assumptions and the references a specialist would expect. Each paper stands on its own, with its own authors, abstract and references, and can be cited independently of the book. A chapter can lean on several papers, and a paper can support several chapters.

No papers have been written yet.

References

- Bekenstein, Jacob D. 1981. “Universal Upper Bound on the Entropy-to-Energy Ratio for Bounded Systems.” *Physical Review D* 23 (2): 287–98. <https://doi.org/10.1103/physrevd.23.287>.
- Bennett, C. H. 1973. “Logical Reversibility of Computation.” *IBM Journal of Research and Development* 17 (6): 525–32. <https://doi.org/10.1147/rd.176.0525>.
- Bérut, Antoine, Artak Arakelyan, Artyom Petrosyan, Sergio Ciliberto, Raoul Dillenschneider, and Eric Lutz. 2012. “Experimental Verification of Landauer’s Principle Linking Information and Thermodynamics.” *Nature* 483 (7388): 187–89. <https://doi.org/10.1038/nature10872>.
- Landauer, R. 1961. “Irreversibility and Heat Generation in the Computing Process.” *IBM Journal of Research and Development* 5 (3): 183–91. <https://doi.org/10.1147/rd.53.0183>.
- Lloyd, Seth. 2000. “Ultimate Physical Limits to Computation.” *Nature* 406 (6799): 1047–54. <https://doi.org/10.1038/35023282>.
- Margolus, Norman, and Lev B. Levitin. 1998. “The Maximum Speed of Dynamical Evolution.” *Physica D: Nonlinear Phenomena* 120 (1-2): 188–95. [https://doi.org/10.1016/s0167-2789\(98\)00054-2](https://doi.org/10.1016/s0167-2789(98)00054-2).
- Turing, A. M. 1937. “On Computable Numbers, with an Application to the Entscheidungsproblem.” *Proceedings of the London Mathematical Society* s2-42 (1): 230–65. <https://doi.org/10.1112/plms/s2-42.1.230>.

A Concepts

Every technical term the book uses, in plain language. Each entry lists the pages that use it.

Arrow of time

The difference between past and future that we observe in everyday life: eggs break but do not unbreak, and we remember yesterday but not tomorrow. It stands out because the fundamental laws of physics barely distinguish between the two directions of time.

Related: [Entropy](#), [Second law of thermodynamics](#)

Bit

The basic unit of information: the answer to one yes-or-no question. A memory that can hold one of two states, such as 0 or 1, stores one bit.

Related: [Landauer's principle](#), [Entropy](#)

Entropy

A measure of how many microscopic arrangements of a system are compatible with what we observe at large scale. The more ways the parts can be arranged without changing how the whole looks, the higher the entropy.

Related: [Second law of thermodynamics](#), [Arrow of time](#), [Bit](#)

Landauer's principle

Erasing one bit of information in surroundings at temperature T releases at least $k_B T \ln 2$ of heat. Forgetting has a minimum physical cost. The bound was proposed by Rolf Landauer in 1961 and confirmed experimentally in 2012.

Related: [Bit](#), [Reversible computing](#), [Maxwell's demon](#)

References: (Landauer 1961; Bérut et al. 2012)

Maxwell's demon

A thought experiment devised by James Clerk Maxwell in 1867: a tiny being opens and closes a door between two boxes of gas, letting fast molecules go one way and slow ones the other. It seems to lower entropy for free. The accepted resolution is that the demon must record what it sees, and erasing that record eventually pays the entropy back.

Related: [Second law of thermodynamics](#), [Landauer's principle](#)

Reversible computing

Computing without erasing information, so that every step can be run backward. Because nothing is erased, Landauer's principle sets no minimum energy cost per operation: in principle, a reversible computer running slowly enough can dissipate as little energy as we like.

Related: [Landauer's principle](#), [Bit](#)

References: (Bennett 1973)

Second law of thermodynamics

In an isolated system, entropy does not decrease over time. It is a statistical law: a decrease is not forbidden, only overwhelmingly unlikely for systems with many parts.

Related: [Entropy](#), [Arrow of time](#), [Maxwell's demon](#)

Turing machine

An abstract model of computation introduced by Alan Turing in 1936: a machine that reads and writes symbols on an unlimited tape, following a finite table of rules. Anything a modern computer can compute, a Turing machine can compute too, given enough time and tape.

Related: [Bit](#)

References: (Turing 1937)

B Notation

Every symbol keeps the same meaning across the whole book.

Symbol	Meaning	Unit
k_B	Boltzmann constant	J K^{-1}
T	Absolute temperature	K
S	Entropy	J K^{-1}
E	Energy	J
$\hbar$	Reduced Planck constant, $h/2\pi$	J s
c	Speed of light in vacuum	m s^{-1}

C Constants

Values from the 2019 revision of the SI, where the constant is exact by definition, and from CODATA otherwise.

Constant	Symbol	Value	Unit	Standard uncertainty	Source
Speed of light in vacuum	c	299792458	m s^{-1}	exact	SI (2019), exact by definition
Planck constant	h	$6.62607015 \times 10^{-34}$	J s	exact	SI (2019), exact by definition
Reduced Planck constant	$\hbar$	$1.054571817 \times 10^{-34}$	J s	exact	$h / 2 \pi$, exact; value truncated to 10 significant digits
Boltzmann constant	k_B	1.380649×10^{-23}	J K^{-1}	exact	SI (2019), exact by definition
Newtonian constant of gravitation	G	6.67430×10^{-11}	$\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$	1.5×10^{-15}	CODATA 2018

D A Short Primer on Statistical Mechanics

This appendix has not been written yet.

E Moderators and Curators

The people who look after the book. Maintainers look after the whole repository. Moderators look after a part: they review the changes to its pages and triage its issues. Curators look after single pages and review their dossiers. People are named by their GitHub handle. How the roles work, and how to take one, is in [GOVERNANCE.md](#).

Maintainers

[JoaoAlisson](#)

Moderators

Part	Moderators
The Machine and the Question	none yet
Information Is Physical	none yet
The Limits of the Machine	none yet
Reversing Entropy	none yet
The Impact on Time	none yet
Building Alan	none yet
Papers	none yet

Curators

No page has a curator yet.